

Ricardo Moura Vargas | Senior Software Engineer

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SUMMARY

I'm a Senior Software Engineer with experience designing, developing and modernizing complex software systems across frontend, full-stack and backend roles. Throughout my career, I've helped teams solve complex technical problems, modernize legacy platforms and deliver scalable, maintainable software.

My strongest expertise is in JavaScript, TypeScript and React, where I've spent the last several years building large-scale web applications, frontend architectures and geospatial visualization platforms. I've also worked across the software stack throughout my career, particularly with .NET/C#, and continue expanding my backend knowledge through personal projects with Python and PHP/WordPress.

Beyond software development, I enjoy technical discovery, software architecture, improving developer experience, and collaborating closely with Product and UX to deliver practical solutions.

I'm particularly interested in applying AI to software engineering. Beyond using AI as a coding tool, I focus on designing AI-assisted software engineering workflows by applying context engineering, structured project knowledge and AI tooling. Combined with years of software engineering experience, this enables me to understand unfamiliar domains quickly, become productive with new technologies and contribute effectively to well-engineered solutions.

CORE STRENGTHS

- Software Architecture & Full-Stack Engineering
- JavaScript, TypeScript & React
- Legacy Modernization
- Scalable Web Platforms
- Design Systems & Component Libraries
- Technical Discovery & Solution Design
- Product & UX Collaboration
- Developer Mentoring
- AI-Assisted Software Engineering

TECHNICAL SKILLS

Frontend

React, JavaScript, TypeScript, HTML, CSS

Backend

.NET/C#, Python, PHP

Database

SQL, Microsoft SQL Server, PostgreSQL, SQLite

State Management & Data Fetching

TanStack Query, Redux, Redux Toolkit, REST APIs, Axios

Frontend Architecture & UI

Web Components, Lit, Storybook, Styled Components, Material UI, Bootstrap, Sass

AI-Assisted Software Engineering

Claude Code, OpenCode, ChatGPT, Claude, Context Engineering, RAG, AI Agents, Custom AI Skills

Mapping & Visualization

OpenLayers, Three.js, WFS, WMS, GeoJSON, ESRI, LiDAR

Testing

Vitest, Jest, React Testing Library

Build Tools & Tooling

Wordpress, Vite, Webpack, ESLint, npm, pnpm, Yarn, Git

Collaboration & Delivery

Scrum, Kanban, Jira, Figma, Bitbucket

LANGUAGES

Portuguese — Native

English — Professional Proficiency

Dutch — Currently Learning

WORK EXPERIENCE

Senior Software Engineer | Feb. 2023 - present | Cyclomedia B.V.

Working across both Cyclomedia's legacy and next-generation geospatial platforms, contributing to feature development, platform modernization and frontend architecture initiatives. Collaborated closely with Product, UX and engineering teams to investigate complex legacy functionality, define technical solutions and implementation strategies, modernize the platform and improve developer experience through scalable and maintainable frontend solutions.

Key Contributions

- Helped build a large-scale TypeScript monorepo consisting of multiple packages and shared libraries, using technologies such as React, Vite, Web Components, Lit, Three.js, OpenLayers, TanStack Query, TanStack Table, Vitest, Storybook, ESLint and Amplitude.
- Participated in the migration of a complex legacy application to a modern architecture, analyzing existing functionality and redesigning solutions to improve maintainability, scalability and developer experience.
- Developed geospatial features by integrating WFS, WMS, GeoJSON and ESRI data sources into OpenLayers-based applications.
- Implemented and maintained OpenLayers-based mapping features, managing multiple geospatial layer types and data sources to deliver interactive map visualization capabilities.
- Implemented spatial data transformations and coordinate system reprojection using different Spatial Reference Systems (SRS), ensuring accurate visualization and interaction across multiple geospatial datasets.
- Conducted a deep analysis of the platform's measurement tools, one of its most critical and technically complex features, producing detailed technical documentation, defining implementation strategies and contributing to the delivery of the new solution.
- Actively participated in feature discovery, brainstorming sessions and UX discussions, influencing product and interface decisions while balancing business requirements, usability and technical feasibility.
- Worked with Three.js, imagery and LiDAR data to develop and maintain interactive panoramic visualization features for large-scale geospatial applications.
- Collaborated on the introduction of AI-assisted software engineering practices, researching and defining context engineering strategies and AI workflows to improve the effectiveness of AI agents during software development.
- Collaborated on the design and organization of project context files, package relationships, development workflows, coding conventions and other structured knowledge used by AI agents during development.

- Improved project documentation by creating architecture documentation, identifying and organizing technical knowledge that enabled more effective retrieval and use by AI-assisted development workflows.
- Designed and implemented an AI-assisted pull request review workflow, defining review rules and coding conventions that allowed developers to use AI feedback as an additional validation layer alongside manual code reviews.
- Collaborated on the integration of CodeGraph through MCP, defining usage guidelines and project context to improve AI navigation and understanding of the codebase.

Senior Frontend Developer | Feb. 2021 to Feb. 2023 | Paraná Banco

Worked on digital lending solutions for public sector employees, contributing to both new product development and the modernization of existing applications. Collaborated closely with UX designers and frontend teams to improve user experience, evolve the company's design system and deliver scalable React-based solutions.

Key Contributions

- Contributed to the development of a new lending platform from the ground up, helping define and implement user-facing features and business workflows.
- Developed and maintained financial applications used in the loan origination and loan portability processes.
- Modernized legacy applications by migrating interfaces to the company's design system and improving consistency across products.
- Contributed to the evolution and maintenance of the frontend component library, helping ensure reusability and consistency across multiple applications.
- Collaborated with UX designers to define new features, refine user journeys and translate business requirements into intuitive user interfaces.
- Participated in code reviews, testing activities and frontend best-practice discussions through the company's Frontend Chapter.
- Worked closely with cross-functional teams during planning, refinement and delivery activities within an Agile/Scrum environment.
- Supported deployment and release activities in homologation and production environments.

Frontend Developer | Jan. 2019 to Feb. 2021 | Agrotis

Worked on Agrotis' grain storage management platform, contributing to new feature development, legacy system modernization and frontend architecture improvements. Helped evolve the application by adopting modern React practices, improving maintainability and supporting the company's design system initiatives.

Key Contributions

- Developed and maintained features for the grain storage module, including the implementation of the services pricing functionality.
- Modernized legacy React applications by migrating class-based components to functional components with Hooks and adopting modern JavaScript (ES6+) practices.
- Contributed to the introduction of TypeScript, helping improve code quality, maintainability and developer experience.
- Collaborated on the adoption and evolution of the company's design system, building reusable UI components and maintaining the shared component library.

- Managed application state using Redux while also simplifying state management approaches when Redux was not required.
- Participated in code reviews and frontend quality initiatives, helping maintain consistency and improve development standards across the application.

Full Stack Developer | Sep. 2014 – Sep. 2015 | PrimeLan Tecnologia

Developed websites following web standards, web applications integrated with Wi-Fi equipment for user authentication, and custom CMS solutions using .NET/C# and PHP. Also developed Vitória Online, the official Android application for the city of Vitória (ES), using Android Studio and Java.

Full Stack Developer | Nov. 2012 – Sep. 2014 | Megawork

Contributed to the development of a product pricing analysis application using .NET/C#, integrated with SAP through RFC. Developed the main dashboard, product management features, supplier listings, and reports using Crystal Reports, while also contributing to the implementation of business rules.

Backend Developer | Feb. 2012 – Nov. 2012 | Phidelis Tecnologia

Worked on a web-based platform for managing schools and universities, built with .NET/C# and SQL Server 2005. Contributed by fixing bugs, developing custom features, reviewing code, and training users on the platform.

Backend Developer | Jul. 2011 – Feb. 2012 | Superius Consultoria LTDA

Contributed to the development of an ERP application as part of a software development team, using .NET/C#, DevExpress, Entity Framework, and SQL Server.

Junior Backend Developer | Jan. 2010 – Jul. 2011 | Vixtime

Contributed to the development of a news management application using .NET/C# and SQL Server, as well as the website for the ES Hoje news outlet. Also developed custom CMS solutions and websites using HTML, CSS, and JavaScript, following W3C web standards.